

VESTA: Time-Budgeted Multimodal Agents for Retail Financial Decision Support

Peri Gunes¹, Ozge Zelal Kucuk², and Harun Benli¹

¹ R&D Center, Infina Software Development Inc., Istanbul, Türkiye
pgunes@infina.com.tr, hbenli@infina.com.tr

² Software Development Faculty, İstanbul Aydın University, Istanbul, Türkiye
ozelalkucuk@stu.aydin.edu.tr

Abstract. In high-volatility emerging markets such as Borsa Istanbul (BIST), retail investors face a low signal-to-noise ratio created by the divergence between unstructured disclosures and rapidly changing charts. Existing decision-support systems process news and technical pictures independently, and they ignore the investor’s temporal bandwidth: the same briefing is delivered whether the user has one minute or ten. This paper presents VESTA, a three-layer agentic system that (i) retrieves portfolio-conditioned textual evidence, (ii) reads candlestick images and the underlying OHLCV numbers, and (iii) spends a declared 1-, 3-, or 10-minute budget by changing the depth of the agentic chain. The contribution is this briefing stack plus an evaluation that treats the user’s minutes as a compute budget and measures leakage, per-tier delivery, and action against buy-and-hold. Text and vision class scores are mixed with GMU, tensor fusion, a scalar gate, mean and concat. A realized-volatility flag on the same day’s OHLCV is recovered at 100% F1 by a closed-form rule, so the headline task is next-day BIST100 direction. On public BIST100 data (2018-2026), no fusion method significantly outperforms GMU (McNemar $p > 0.46$ on seed 0). Coverage, declared latency, and length-normalized information noise vary across the three time tiers. A paper-trading overlay on seed-0 calls does not beat buy-and-hold in the 2025-2026 test window. Source: <https://github.com/ozge-devops/congress>.

Keywords: Multimodal AI · Agentic RAG · Financial decision support · Borsa Istanbul · Temporal bandwidth · Evaluation protocol

1 Introduction

Retail investors sit between the textual density of the Public Disclosure Platform (KAP) [18] and the visual flux of technical charts. The difficulty is not only volume but modality mismatch: a constructive disclosure can coincide with a failed breakout, and the two streams must be reconciled before a decision. Conventional DSS pipelines keep NLP and computer vision in separate pipelines. A second failure mode is temporal: systems emit the same information density whether the user has sixty seconds or ten minutes [7, 17, 22]. On a mobile-first retail platform that mismatch adds intrinsic load [23].

Recent LLM trading agents attack related problems from a different angle. FinMem layers memory by information half-life [27]. VESTA scores a briefing for a retail user whose clock is 1, 3 or 10 minutes: KAP list text and a 40-bar chart. Evaluation reports leakage, per-tier delivery, and action against buy-and-hold. If the user has about one minute, the system emits a short alert or stays silent. If they have about three minutes, it adds one retrieval step over similar past days. If they have about ten minutes, it expands into a briefing with a macro explanation and a hedge sketch. The fusion rule does not change with the clock; only how many agent steps run and how much text is shown. We measure, for each setting, how often a call is issued, how long the pipeline is allowed to run, how many tokens are delivered, and what fraction of that payload is non-actionable. Decision support is scored on next-day BIST direction, not on a realized-volatility flag that can be read off the same day’s OHLCV table. Whenever a chart encoder is used, a numeric OHLCV baseline is reported beside it [11]. The public tables mix unimodal class scores with GMU [2], tensor fusion [28] and a scalar gate. Actionability is a long/short overlay against buy-and-hold, not classification F1 alone. The public chart is bars $t - 40, \dots, t - 1$ and does not contain today’s candle.

2 Related Work

2.1 LLMs and agentic workflows in finance

Financial NLP has moved from sentiment scoring [20] to domain LLMs such as FinGPT [25]. Hallucination on prices and terminology remains a documented failure mode [13], which is why VESTA grounds text in retrieval [15]. Tool-using agents [26] and trading-oriented systems with layered memory [27] are the closest architectural relatives. FinMem allocates memory by the staleness of facts; VESTA allocates compute by the staleness of the user’s attention. FinMem-style agents optimize for the trader; VESTA is scored as a briefing engine for a time-constrained retail user.

2.2 Multimodal fusion and chart understanding

GMU learns a multiplicative gate between modalities [2]; Tensor Fusion Networks model unimodal and bimodal interactions via an outer product [28]; MulT uses directional pairwise cross-modal attention [24]. Jiang and Ji add a forget-gate over cross-modal attention maps [12]. Public tables mix unimodal class scores with GMU, TFN, a scalar gate, mean, concat and a MulT-style layer.

On the vision side, Pix2Struct and MatCha pretrain screenshot-to-text models that read plots [14, 16]. In asset pricing, Jiang, Kelly and Xiu show that CNNs on price-trend images extract return-predictive patterns [11]. That result motivates VisualClaw and requires a baseline that sees the same series as numbers. Without the tabular arm, a vision score on a label computed from OHLCV is uninterpretable.

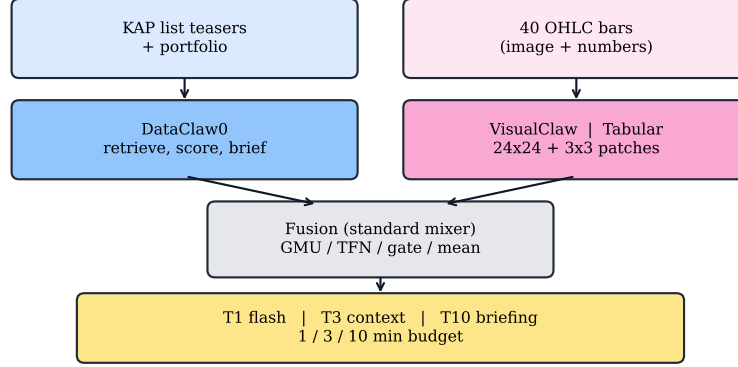


Fig. 1. KAP list text and a 40-bar chart go through a standard mixer; T1, T3 or T10 spends the 1-/3-/10-minute budget.

BIST100 forecasting from technical indicators is studied by Akturk et al. [1]; their features are numeric, which is the required control whenever a chart encoder is scored.

2.3 Cognitive load and adaptive interfaces

Cognitive load theory requires that presented density match working-memory capacity [22, 23, 17]. Information overload is a long-standing MIS construct [7]. Adaptive GUIs have a documented evaluation discipline: measure whether adaptation helps the task, not only whether the widget changed [9, 8]. Summarization evaluation that ignores output length systematically rewards brevity [21]. On Borsa Istanbul the load is concrete: a KAP list, a candlestick, and a clock between a filing and a retail order. Those four literatures set the metrics (Sect. 4.3) and the 1-/3-/10-minute protocol (Sect. 6).

3 System

VESTA has three layers (Fig. 1): perception by two expert agents, an off-the-shelf fusion block, and a temporal synthesis layer that spends the user’s time budget.

3.1 DataClaw0: portfolio-conditioned retrieval

DataClaw0 is a retrieve-score loop [15, 26] (Fig. 2):

1. Embed the day’s KAP filings (and any news teasers) together with the user portfolio P using M3-Embedding [3].

Table 1. Example KAP list teasers in VESTA-Public (subject plus short summary from kap.org.tr). Polarity is the silver codebook.

Date	Ticker	KAP list teaser	Polar.
2018-06-14	BIMAS	Payların Geri Alınmasına İlişkin Bildirim. 13 Haziran 2018 Tarihli Pay Geri Alım İşlemleri	bullish
2018-06-19	ASELS	BISTECH Pay Piyasası Alım Satım Sistemi Duyurusu. Temettü Ödemesi	bullish
2018-05-25	YKBNK	Özel Durum Açıklaması (Genel). Bankamıza verilen idari para cezasının iptaline ilişkin açılan dava hk.	bearish
2020-08-11	FROTO	Özel Durum Açıklaması (Genel). Rekabet Kurulu Soruşturma Kararı Hakkında	bearish

2. Score each document d_i by cosine to P plus the day’s gold, USD/TRY and session moves,

$$R(d_i) = \sigma(\alpha \cos(v_{d_i}, v_P) + \beta_{Au} \Delta_{Au} + \beta_{FX} \Delta_{FX} + \beta_r r_t), \quad (1)$$

with σ the logistic function and Δ_{Au} , Δ_{FX} , r_t the gold, USD/TRY and session simple returns. Cosine on M3 vectors is dense retrieval [3]; Eq. (1) is our score, not the RAG generative likelihood. Public mixers use the 16-d macro+KAP bag; α and the β coefficients are not fitted in Table 4.

3. Keep the top-ranked filings as the text brief that later enters fusion.
4. Stop after this pass on a one-minute budget (T1). On a three- or ten-minute budget, run one more retrieval hop over historical analogues, then write.

The text dataset is VESTA-Public: one row per ticker-day (37,046 events; 27 BIST names plus XU100; 24 May 2018 to 19 August 2026). A row is kept if the name moved at least 2%, the vol flag fired, RSI was extreme, or a KAP list item was published that calendar day. 19,645 rows have a same-day KAP teaser; 15,212 have a cached public HTML body. Yahoo headlines exist for 41 rows. Table 1 shows four KAP list strings. We do three things with them: (i) a Turkish/English lexicon plus subject priors (dividend and buyback tilt bullish; fine and probe tilt bearish) yields kap_polarity; (ii) the same strings, plus USD/TRY, gold and the session, become an eight-plus-eight feature vector for the mixers; (iii) MiniLM [19] and BGE-M3 embed the concatenated list text. The templated brief a T10 user would read looks like: “2018-06-14 BIMAS.IS. Session return +0.12%. USD/TRY . . . Brief polarity bullish; chart cue none. Payların Geri Alınmasına İlişkin Bildirim. 13 Haziran 2018 Tarihli Pay Geri Alım İşlemleri.”

3.2 VisualClaw: candles, pixels, tokens

VisualClaw draws the chart from OHLC bars. Figure 3 is one XU100 window from the test split.

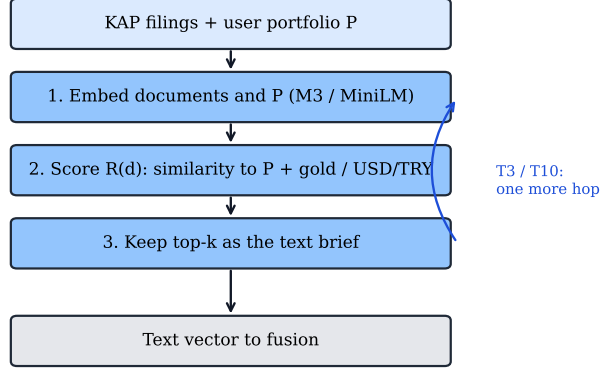


Fig. 2. DataClaw0 retrieve-score loop. The blue hop is a second retrieval, used only on a T3 or T10 budget.

1. Take the 40 sessions strictly before day t (open, high, low, close). Today's candle is not in the picture.
2. Paint a grayscale candlestick: a one-pixel wick and a rectangular body. Close $\geq$ open is dark (30/255); close $<$ open is light (180/255). Price is min-max scaled to that window.
3. Resize to 24×24 . The vision MLP flattens this to a 576-d vector and classifies with hidden layers 64-32. That is the vision row in Table 4.
4. For the MulT-style mixer only, cut the same 24×24 image into a 3×3 grid of 8×8 patches (nine visual tokens). The 16-d text brief is split into two tokens. Projections are frozen (seed 0); only the MLP on top is trained.

Two extra probes sit beside the mixer: a frozen ImageNet ViT-B/16 [6] that resizes a colour chart to 224×224 and uses that model's own 16×16 patches; and zero-shot DePlot / MatCha, which see a matplotlib RGB screenshot of the same 40 bars (Fig. 3a). Tabular OHLCV at day t is always reported beside the picture [11]. Public tables train the flattened 24×24 MLP; they do not add a recency term to patch positions.

3.3 Fusion

Arevalo et al. write the bimodal GMU as [2]

$$\begin{aligned} h_v &= \tanh(W_v x_v), & h_t &= \tanh(W_t x_t), \\ z &= \sigma(W_z [x_v; x_t]), & h &= z h_v + (1 - z) h_t, \end{aligned} \quad (2)$$

so z near 1 keeps the visual branch. Zadeh et al. flatten $[h_t; 1] \otimes [h_v; 1]$ (TFN) [28]. In Table 4, x_t and x_v are unimodal class-score vectors: W_v , W_t and W_z are unused. The GMU row keeps Arevalo's mix with z a sigmoid of the score gap.

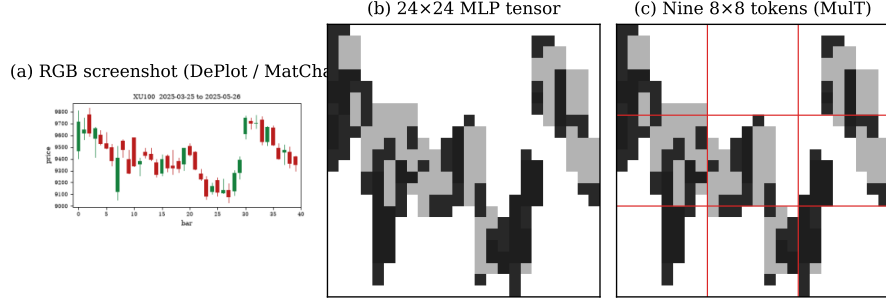


Fig. 3. VisualClaw on one XU100 test window (40 bars before 27 May 2025). (a) RGB screenshot for DePlot / MatCha. (b) 24×24 grayscale tensor the MLP flattens. (c) Nine 8×8 tokens for the MulT-style mixer.

The TFN row uses that outer product on the same scores. The scalar-gate row uses

$$g = \sigma(4(p_{\text{text}} - p_{\text{vision}})), \quad (4)$$

$$h = g \tanh(p_{\text{text}}) + (1 - g) \tanh(p_{\text{vision}}), \quad (5)$$

so g near 1 keeps text. A small MLP reads the mix. Mean and concat are unweighted. The MulT-style row is one CPU attention layer on two text tokens and nine 8×8 patches, with frozen projections [24]. The fused call drives T1 / T3 / T10.

3.4 Temporal Orchestration Layer

Given fused confidence $|g - 0.5|$ and a user budget $B \in \{1, 3, 10\}$ minutes, the layer selects a chain:

- T1 (flash). Emit a gated alert only if $|g - 0.5| > \tau$ with $\tau = 0.18$; otherwise abstain. Target latency < 1 s.
- T3 (context). Always emit the fused call plus one retrieval hop over historical analogues.
- T10 (briefing). Expand into a macro explanation and a hedge sketch (e.g. airline exposure versus a gold certificate), at several seconds of wall-clock time.

Chain depth changes with B ; the mixer stays fixed. The same fused call is spent as a flash, a short context, or a briefing. Section 5.4 reports coverage, latency, tokens and noise per tier.

Table 2. Proxy accuracy vs next-day direction y_{t+1} (test $n = 308$). Macro/KAP are single-split rules; vision/tabular are mean $\pm$ 95% CI over five seeds.

Proxy	Acc. (%)
Macro flags (USD/TRY, gold, vol, shock)	54.9
KAP polarity (lexicon on public list text)	49.7
Vision 24×24 MLP	52.9 ± 2.3
Tabular OHLCV	50.6 ± 0.0

4 Evaluation Protocol

4.1 Diagnostic: the realized-volatility label

The 2σ realized-vol flag at day t is 20-day realized vol (annualized) above the 60-day rolling mean plus 2σ of that 20-day vol series, all taken from the tabular vector at t (100% F1). The public chart does not contain today’s candle, so vision F1 on this label is lag-1 recovery. We report the flag only as a diagnostic.

4.2 Primary task: forward direction

The primary label at close of day t is the next session’s direction: $y_{t+1} = 1$ if the BIST100 simple return r_{t+1} is positive, and $y_{t+1} = 0$ otherwise. The label is unknown at the close of day t . We also report a five-session horizon. Splits are chronological (70/15/15) to block look-ahead [1].

4.3 Metrics

Classification. Accuracy and macro-F1, mean $\pm$ 1.96 standard error over five seeds. McNemar with continuity correction [5] on seed 0 of one chronological holdout. Table 4 has thirteen rows; McNemar is mixer-only. With that many rows we follow Harvey et al. [10] and do not pick a winner after an uncorrected search.

Sentiment / technical accuracy. Table 2 reports proxy accuracy versus next-day direction. Macro flags and KAP polarity are deterministic rules on one chronological split (no seed CI). Vision and tabular are five-seed means from Table 4. Three KAP codebooks on the 10k slice give Fleiss $\kappa = 0.50$; chart A/B Cohen $\kappa = 0.52$ [4]. The 250-row annotator columns are codebook B.

Information noise. This is the computational cognitive measure. Let N_{raw} be tokens in the unprocessed window (the raw KAP list plus session flags) and N_{del} tokens actually delivered in the T1, T3 or T10 briefing. Let N_{irrel} be delivered tokens that are not in a relevance lexicon for that window (session, USD/TRY, gold, vol, KAP digest). The index $\text{IN}_{\text{old}} = N_{\text{irrel}}/N_{\text{raw}}$ rewards short answers: a 40-token briefing that is 80% filler still looks clean against a 700-token feed [21]. We therefore report

$$\text{IN}_{\text{new}} = N_{\text{irrel}}/N_{\text{del}}, \quad (6)$$

the fraction of the delivered finance briefing that is non-actionable, together with the compression ratio $N_{\text{del}}/N_{\text{raw}}$. IN_{new} is our index; Sun et al. motivate dividing by delivered length. Both counts use templated token bags with filler padding. Table 6 is that computational proxy; human NASA-TLX is Sect. 6.

Decision quality. Classification F1 does not say whether acting on the output helps [8]. We paper-trade a long/short overlay: +1 if the model predicts up, −1 otherwise, with 5 bp round-trip cost, and compare Sharpe, total return and max drawdown to buy-and-hold. Table 7 is that control; Sect. 6 is the human cognitive protocol.

4.4 Public replication corpus

Numbers below use VESTA-Public: XU100.IS, USDTRY and COMEX gold from 24 May 2018 to 19 August 2026 (2,052 index windows after a 40-day lookback; test $n = 308$ from 27 May 2025), 27 BIST names, and 39,890 KAP list filings from kap.org.tr. Tables 3 to 7 are index-level; the 37,046-row event panel also labels constituents. KAP-only polarity: 7,172 / 3,742 / 26,132 (bullish / bearish / neutral); 15,212 rows have a cached public HTML body. Mixers use the 16-d macro+KAP vector; MiniLM and BGE-M3 are unimodal text probes on the same list teasers.

5 Experiments

Unimodal encoders are sklearn MLPs (64-32, Adam, early stopping) on standardized features. Vision sees a 24×24 grayscale candlestick of bars $t - 40, \dots, t - 1$; tabular sees eleven OHLCV/macro features at t , including the ingredients of the leaked vol rule; text sees the sixteen-dimensional macro+KAP vector. Five seeds $\{0, \dots, 4\}$. Training is on CPU. Public BGE-M3 and frozen ViT-B/16 are unimodal probes; they remain unimodal in Table 4.

5.1 Diagnostic: the leaked volatility label

Table 3 and Fig. 4 confirm the closed-form critique. The 2σ rule on the tabular vector scores 100% F1. A tabular MLP that is given those numbers reaches 96.8% accuracy (macro-F1 86.6%; the F1 drop is class imbalance: 25 positives among 308 test days, 8.1%; 9.4% of all 2,052 windows). The 24×24 vision encoder, which never sees day t 's candle, falls to 90.6% accuracy and 58.2% F1.

5.2 Primary task: next-day direction

Table 4 reports the leakage-free task (test prior 52.3% up-days). Text-only (macro+KAP) macro-F1 is 47.3 ± 1.5 against a majority F1 of 34.3; MiniLM 53.0 ± 0.4 / 50.7 ± 1.9 ; public BGE-M3 51.1 ± 1.9 / 50.2 ± 2.1 ; frozen ViT-B/16 50.9 ± 1.4 / 50.3 ± 1.6 . Mixers share 16-d text and the 24×24 image; none beats GMU on seed 0 (min. McNemar $p = 0.46$). Mean fusion is the five-seed

Table 3. Diagnostic task: 2σ realized-vol label, computable from the tabular vector at t . Mean $\pm$ 95% CI over five seeds. Vision sees bars $t - 40, \dots, t - 1$ only.

Method	Accuracy (%)	Macro-F1 (%)
Closed-form OHLCV rule	100.0	100.0
Tabular MLP (same numbers)	96.8 ± 0.0	86.6 ± 0.0
Vision MLP (chart image)	90.6 ± 1.1	58.2 ± 3.0

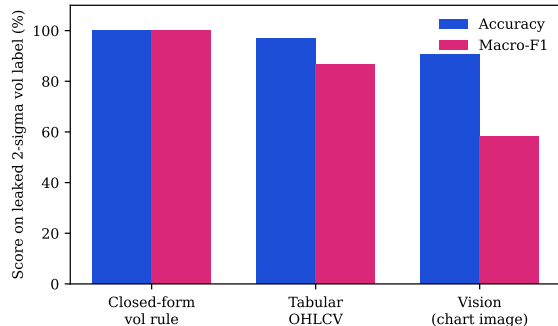


Fig. 4. Diagnostic 2σ -vol label. The closed-form OHLCV rule scores 100% F1; the chart omits day t .

point estimate (53.0 ± 2.2 / 52.4 ± 2.3). Five-day tabular accuracy is 59.1% with collapsed F1 (39.9%).

Replacing the gate with a fixed mean changes five-seed F1 from 51.4 ± 1.8 to 52.4 ± 2.3 (seed-0 McNemar $p = 1.00$).

5.3 Zero-shot Pix2Struct and MatCha

We run zero-shot DePlot (Pix2Struct chart-to-table) [14] and MatCha-ChartQA [16] on CPU matplotlib screenshots of the full test window ($n = 308$), with no fine-tune. Calls are $\text{sign}(\hat{y}_{\text{last}} - \hat{y}_{\text{first}})$. Table 5 also scores the numeric window trend of the input OHLC, which is still not y_{t+1} . DePlot parses 307/308 images and recovers that trend on 77.9% of days; next-day accuracy is 52.9%. MatCha-ChartQA often returns the bar index (parse rate 166/308) and matches the majority class on y_{t+1} .

5.4 Per-tier measurement

Table 6 and Fig. 6 measure the Temporal Orchestration Layer. Latencies 0.9 / 2.1 / 3.4s are declared budgets, not wall-clock. On seed 0, T1 covers 32.5% of days (60.0% among emitted calls; 55.2% after majority fill). T3 and T10 always emit (coverage 100%); accuracy among those calls is 49.7%. Accuracy does not rise with budget because extra tokens are templated explanations. $\text{IN}_{\text{old}} < 7\%$ for every tier; IN_{new} is 60.3 / 82.1 / 93.6% on filler-padded templates.

Table 4. Next-day BIST100 direction on VESTA-Public (test $n = 308$). Mean ± 1.96 se, five seeds. Rows are ordered by macro-F1. MiniLM, BGE-M3 and ViT are unimodal probes. Mixers: GMU [2], TFN [28], MulT-style [24]. GMU/TFN/gate mix unimodal class scores; Arevalo’s W_v, W_t, W_z are unused.

Configuration	Accuracy (%)	Macro-F1 (%)
Mean fusion	53.0 ± 2.2	52.4 ± 2.3
TFN	52.1 ± 2.1	51.7 ± 2.1
Vision-only (24×24 MLP)	52.9 ± 2.3	51.4 ± 2.6
Scalar gate (score-space)	51.8 ± 1.7	51.4 ± 1.8
Concatenation	52.8 ± 1.8	51.1 ± 2.3
Text-only (MiniLM KAP)	53.0 ± 0.4	50.7 ± 1.9
GMU	50.6 ± 2.4	50.6 ± 2.4
Vision-only (frozen ViT-B/16)	50.9 ± 1.4	50.3 ± 1.6
Text-only (public BGE-M3)	51.1 ± 1.9	50.2 ± 2.1
MulT-style (CPU, one layer)	50.2 ± 3.1	48.4 ± 3.0
Text-only (macro + KAP)	52.1 ± 2.4	47.3 ± 1.5
Tabular OHLCV	50.6 ± 0.0	41.8 ± 0.0
Majority class	52.3 ± 0.0	34.3 ± 0.0

Table 5. Zero-shot CPU screenshot VLMs on the full test window ($n = 308$), with no fine-tune.

Method	Acc. vs y_{t+1}	Macro-F1	Acc. vs window trend
Majority	52.3	34.3	n/a
Numeric window trend	51.6	49.5	100.0
DePlot (Pix2Struct)	52.9	51.9	77.9
MatCha-ChartQA	52.3	46.0	58.4

5.5 Decision quality

Table 7 asks whether acting on the output would have helped. In the May 2025 to August 2026 test window BIST100 buy-and-hold returned +56.3% (Sharpe 1.69). Every seed-0 overlay we tried underperformed that long-only benchmark after 5 bp costs. Vision-only (Sharpe 0.37) and the scalar gate (0.33) were positive but far below BH; GMU and tabular were negative. A bull-market holdout is a hard test for a long/short overlay [10].

6 Cognitive protocol

The cognitive work has two layers. The first is computational and is already in Table 6. A BIST retail user on a mobile desk has about one, three or ten minutes between a KAP drop and an order. That clock is the independent variable. The dependent quantities are coverage (how often the system speaks), declared latency, tokens delivered, and IN_{new} (what fraction of the delivered briefing is non-actionable). The token bags are built from the day’s finance flags: session

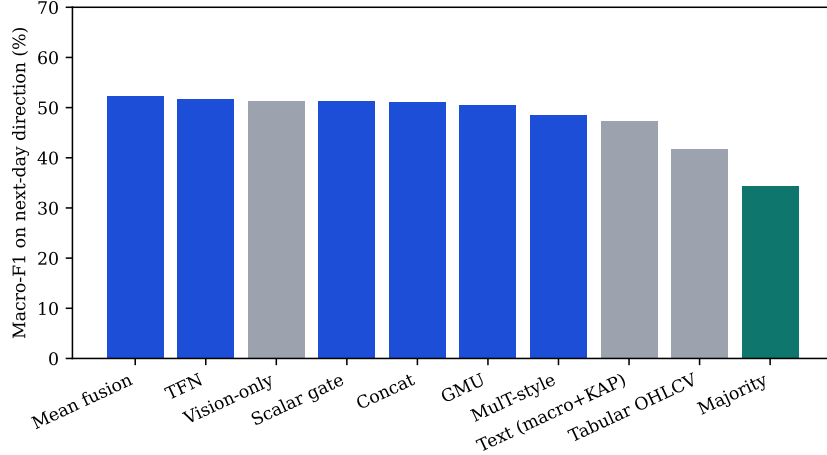


Fig. 5. Macro-F1 for mixers and sklearn unimodal rows of Table 4. MiniLM, BGE-M3 and ViT are in the table. Mean $\pm$ 95% CI is in the table.

Table 6. Temporal Orchestration Layer on seed 0. Latency is a declared budget, not measured GPU time. Acc. is among emitted calls; T1 overall accuracy with majority fill is 55.2%.

Tier	Cov. (%)	Lat. (s)	Acc. (%)	Tokens	IN _{old}	IN _{new}
T1 flash	32.5	0.9	60.0	6	0.5%	60.3%
T3 context	100.0	2.1	49.7	17	1.9%	82.1%
T10 briefing	100.0	3.4	49.7	52	6.8%	93.6%

return, USD/TRY, gold, realized vol, a KAP digest, and, on T3/T10, a historical analogue and a hedge sketch. Those counts measure load on the payload. They do not measure load on a person [9, 8].

The second layer is a human protocol in **study/**. Twelve retail-account holders, within-subject, see eight sealed snapshots drawn from the public test window of VESTA-Public (two XU100 sessions and six constituents: SASA, SISE, YKBNK, TAVHL, AKBK, PETKM). Each snapshot is a real calendar day. The data that arrive on screen are the session return, the USD/TRY and gold moves, RSI, and the KAP list teasers published that day. The sealed gold is the subsequent week’s realized direction of that name. Participants see one snapshot at a time under a 1-, 3- or 10-minute cap, in one of three information conditions: the raw KAP+OHLC feed, a unimodal text or chart, or the VESTA tier that matches the cap. Order is Latin-squared. Outcomes are the week-direction call, time-to-decision in seconds, six NASA-TLX items on a 0-20 scale (mental demand, physical demand, time pressure, perceived success, effort, frustration), and a 7-point trust item (“I would act on this briefing with real money”). Consent and the bilingual instrument are in **study/**.

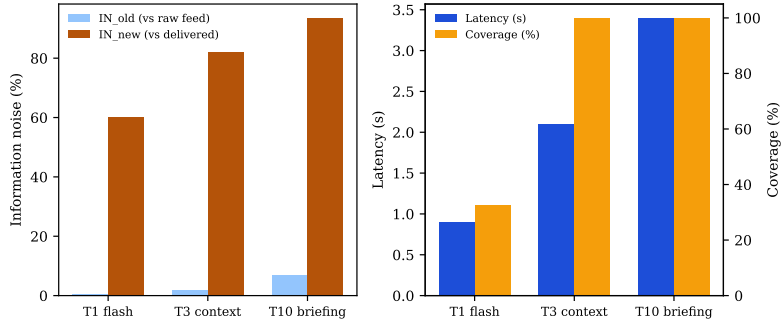


Fig. 6. Left: length-normalized noise of the delivered briefing (IN_{old} vs IN_{new}). Right: declared latency and coverage for T1, T3 and T10.

Table 7. Paper-trading overlay on seed-0 calls, chronological test window, 5 bp costs. BH is long-only BIST100.

Policy	Hit (%)	Sharpe	Total ret.	Max DD
Buy-and-hold	52.3	1.69	+56.3%	14.5%
Vision-only overlay	50.0	0.37	+7.6%	16.4%
Scalar-gate overlay	49.7	0.33	+6.3%	17.1%
Tabular overlay	50.6	-0.11	-6.2%	23.0%
GMU overlay	48.1	-0.74	-21.8%	27.4%

Responses have not been collected. A model dry-run on the sealed gold hits 4/8 name-level weeks for the raw session sign; index mixers are scored only on the two XU100 weeks (text 0/2, gated 1/2). NASA-TLX and trust stay blank.

7 Discussion and limitations

VESTA’s contribution is a time-budgeted briefing for BIST retail users and the evaluation that scores it: time as a budget, leakage made visible, delivery measured per tier, and action compared with buy-and-hold. DataClaw0, VisualClaw and gated delivery are the engineering stack. Next-day accuracy is modest and the overlay loses to buy-and-hold on a bull holdout. The time budget shows up in the tier table: T1 stays silent on most days; T3 and T10 always fire and add templated tokens, so IN_{new} rises with depth.

Public BGE-M3 is run on KAP list teasers; full disclosure-detail pages are not in the package. Frozen ViT-B/16 does not beat the 24×24 MLP. DePlot and MatCha are zero-shot on 308 days. The 12-participant NASA-TLX study has no responses. The public chart uses bars $t - 40, \dots, t - 1$. Information noise and tier latencies are simulated. Table 4 mixers fuse unimodal class scores; Arevalo’s W_v, W_t, W_z are unused.

8 Conclusion

This paper’s contribution is a briefing stack that gives a BIST retail user KAP list text and a 40-bar candlestick under a 1-, 3- or 10-minute budget, together with an evaluation that treats that clock as a compute budget and measures leakage, per-tier delivery, and action against buy-and-hold. Financial LLM agents such as FinMem organise memory by how old a fact is and are scored as traders [27]. Chart-reading and fusion papers typically introduce a new mixer or report a high F1 on a label computed from the same day’s OHLCV. Conventional decision-support systems keep text and charts in separate pipelines and emit the same density whether the user has one minute or ten [7, 22]. DataClaw0 retrieves KAP list text, VisualClaw reads the candlestick, a standard gate fuses the two streams, and the Temporal Orchestration Layer spends the budget by changing chain depth. The protocol scores next-day BIST100 direction, places a tabular OHLCV baseline beside every chart encoder, measures coverage, declared latency, tokens and length-normalized noise per tier, and paper-trades a long/short overlay against buy-and-hold. VESTA-Public (37,046 events, KAP list teasers, chronological splits) makes those numbers reproducible.

On the public test window ($n = 308$) the benefit is the protocol itself: leakage is visible, tiers are measured, and action is scored against buy-and-hold. A 2σ realized-vol flag is recovered at 100% F1 by a closed-form OHLCV rule, so the headline label is next-day direction. No mixer beats GMU on that task (mean fusion is the point estimate, 53.0 ± 2.2 accuracy / 52.4 ± 2.3 macro-F1; seed-0 McNemar $p > 0.46$). A long/short overlay loses to buy-and-hold in the 2025-2026 window (BH Sharpe 1.69 versus at most 0.37). What the time budget changes is coverage (T1 32.5% on seed 0), declared latency, tokens, and IN_{new} (60.3 / 82.1 / 93.6% on templated bags). Those four measurements are what the protocol adds: time as a budget, leakage made visible, delivery scored per tier, and action compared with buy-and-hold.

Three items remain open. The computational cognitive measurement is Table 6; the 12-participant NASA-TLX study in `study/` has no responses, so human load and trust are unspecified. There is no three-annotator gold set; polarity is silver codebook output (Fleiss $\kappa = 0.50$ on the 10k slice). Public text is KAP list teasers; full disclosure-detail pages are absent. Table 4 mixers fuse unimodal class scores; Arevalo’s W_v, W_t, W_z are unused. The public chart is bars $t-40, \dots, t-1$, IN and tier latencies are simulated, and DePlot / MatCha remain zero-shot on 308 screenshots. Those are the next measurements.

Disclosure of Interests. The authors declare that they have no competing interests relevant to the content of this article. P.G. and H.B. are employed by Infina Software Development Inc., which operates a BIST retail platform; the public replication does not use proprietary KAP text.

References

1. Akturk, Z., Kılıç, E., Algorabi, Ö., Ulu, M., Türkan, Y.S., Namlı, E.: Real-time forecast of BIST100 index under market volatility and uncertainty. *Computational Economics* **67**(4), 3249–3274 (2026). <https://doi.org/10.1007/s10614-025-10986-1>
2. Arevalo, J., Solorio, T., y Gómez, M.M., González, F.A.: Gated multi-modal units for information fusion. *arXiv preprint arXiv:1702.01992* (2017). <https://doi.org/10.48550/arXiv.1702.01992>
3. Chen, J., Xiao, S., Zhang, P., Luo, K., Lian, D., Liu, Z.: M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In: *Findings of the Association for Computational Linguistics: ACL 2024*. pp. 2318–2335 (2024). <https://doi.org/10.18653/v1/2024.findings-acl.137>
4. Cohen, J.: A coefficient of agreement for nominal scales. *Educational and Psychological Measurement* **20**(1), 37–46 (1960). <https://doi.org/10.1177/001316446002000104>
5. Dietterich, T.G.: Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation* **10**(7), 1895–1923 (1998). <https://doi.org/10.1162/089976698300017197>
6. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., Houlsby, N.: An image is worth 16x16 words: Transformers for image recognition at scale. In: *International Conference on Learning Representations* (2021), [arXiv:2010.11929](https://arxiv.org/abs/2010.11929)
7. Eppler, M.J., Mengis, J.: The concept of information overload: A review of literature from organization science, accounting, marketing, MIS, and related disciplines. *The Information Society* **20**(5), 325–344 (2004). <https://doi.org/10.1080/01972240490507974>
8. Findlater, L., Gajos, K.Z.: Design space and evaluation challenges of adaptive graphical user interfaces. *AI Magazine* **30**(4), 68–73 (2009). <https://doi.org/10.1609/aimag.v30i4.2268>
9. Gajos, K.Z., Czerwinski, M., Tan, D.S., Weld, D.S.: Exploring the design space for adaptive graphical user interfaces. In: *Proceedings of the Working Conference on Advanced Visual Interfaces (AVI '06)*. pp. 201–208 (2006). <https://doi.org/10.1145/1133265.1133306>
10. Harvey, C.R., Liu, Y., Zhu, H.: ... and the cross-section of expected returns. *The Review of Financial Studies* **29**(1), 5–68 (2016). <https://doi.org/10.1093/rfs/hhv059>
11. Jiang, J., Kelly, B., Xiu, D.: (Re-)Imag(in)ing price trends. *The Journal of Finance* **78**(6), 3193–3249 (2023). <https://doi.org/10.1111/jofi.13268>
12. Jiang, M., Ji, S.: Cross-modality gated attention fusion for multi-modal sentiment analysis. *arXiv preprint arXiv:2208.11893* (2022). <https://doi.org/10.48550/arXiv.2208.11893>
13. Kang, H., Liu, X.Y.: Deficiency of large language models in finance: An empirical examination of hallucination. In: *I Can’t Believe It’s Not Better Workshop: Failure Modes in the Age of Foundation Models* (2023), *neurIPS 2023 workshop*. <https://openreview.net/forum?id=SGiQxu8zFL>. [arXiv:2311.15548](https://arxiv.org/abs/2311.15548)
14. Lee, K., Joshi, M., Turc, I.R., Hu, H., Liu, F., Eisenschlos, J.M., Khandelwal, U., Shaw, P., Chang, M.W., Toutanova, K.: Pix2Struct: Screenshot parsing as pretraining for visual language understanding. In: *Proceedings of the 40th International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 202, pp. 18893–18912 (2023)

15. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., tau Yih, W., Rocktäschel, T., Riedel, S., Kiela, D.: Retrieval-augmented generation for knowledge-intensive NLP tasks. In: *Advances in Neural Information Processing Systems*. vol. 33, pp. 9459–9474 (2020)
16. Liu, F., Piccinno, F., Krichene, S., Pang, C., Lee, K., Joshi, M., Altun, Y., Collier, N., Eisenschlos, J.: MatCha: Enhancing visual language pretraining with math reasoning and chart derendering. In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. pp. 12756–12770 (2023). <https://doi.org/10.18653/v1/2023.acl-long.714>
17. Miller, G.A.: The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review* **63**(2), 81–97 (1956). <https://doi.org/10.1037/h0043158>
18. Public Disclosure Platform (KAP): KAP public disclosure platform. <https://www.kap.org.tr> (2026), accessed 27 August 2026
19. Reimers, N., Gurevych, I.: Sentence-BERT: Sentence embeddings using siamese BERT-networks. In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. pp. 3982–3992 (2019). <https://doi.org/10.18653/v1/D19-1410>
20. Sohagir, S., Wang, D., Pomeranets, A., Khoshgoftaar, T.M.: Big data: Deep learning for financial sentiment analysis. *Journal of Big Data* **5**(1), 3 (2018). <https://doi.org/10.1186/s40537-017-0111-6>
21. Sun, S., Shapira, O., Dagan, I., Nenkova, A.: How to compare summarizers without target length? pitfalls, solutions and re-examination of the neural summarization literature. In: *Proceedings of the Workshop on Methods for Optimizing and Evaluating Neural Language Generation*. pp. 21–29 (2019). <https://doi.org/10.18653/v1/W19-2303>
22. Sweller, J.: Cognitive load during problem solving: Effects on learning. *Cognitive Science* **12**(2), 257–285 (1988). https://doi.org/10.1207/s15516709cog1202_4
23. Sweller, J.: The development of cognitive load theory: Replication crises and incorporation of other theories can lead to theory expansion. *Educational Psychology Review* **35**(4), 95 (2023). <https://doi.org/10.1007/s10648-023-09817-2>
24. Tsai, Y.H.H., Bai, S., Liang, P.P., Kolter, J.Z., Morency, L.P., Salakhutdinov, R.: Multimodal transformer for unaligned multimodal language sequences. In: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. pp. 6558–6569 (2019). <https://doi.org/10.18653/v1/P19-1656>
25. Yang, H., Liu, X.Y., Wang, C.D.: FinGPT: Open-source financial large language models. *arXiv preprint arXiv:2306.06031* (2023). <https://doi.org/10.48550/arXiv.2306.06031>
26. Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., Cao, Y.: ReAct: Synergizing reasoning and acting in language models. In: *International Conference on Learning Representations* (2023), [arXiv:2210.03629](https://arxiv.org/abs/2210.03629)
27. Yu, Y., Li, H., Chen, Z., Jiang, Y., Li, Y., Zhang, D., Liu, R., Suchow, J.W., Khashanah, K.: FinMem: A performance-enhanced LLM trading agent with layered memory and character design. *Proceedings of the AAAI Symposium Series* **3**(1), 595–597 (2024). <https://doi.org/10.1609/aaais.v3i1.31290>
28. Zadeh, A., Chen, M., Poria, S., Cambria, E., Morency, L.P.: Tensor fusion network for multimodal sentiment analysis. In: *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*. pp. 1103–1114 (2017). <https://doi.org/10.18653/v1/D17-1115>